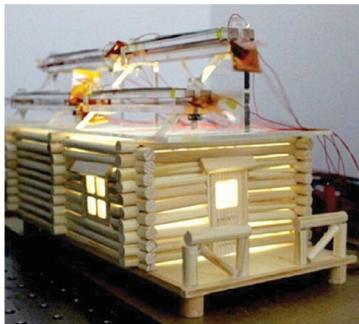


Nanogenerator for harvesting wind, solar energy to power IoT devices



The Internet of Things (IoT) could make cities smarter by connecting an extensive network of tiny communications devices to make life more efficient. But all such machines require a lot of energy. Rather than adding to the global reliance on fossil fuels to power the network, researchers have come up with a solution.

For the first time, researchers have integrated two energy-harvesting technologies in one: a silicon solar cell and a nanogenerator that can convert wind energy into electrical output. The solar cell component of the system delivers eight milliwatts of power output (one milliwatt can light up 100 small LEDs). The wind harvesting component delivers up to 26 milliwatts. Together, under simulated sun and wind conditions, four devices on the roof of a model home could turn on the LEDs inside and power a temperature-humidity sensor. Installed in large numbers on rooftops, the hybrid device could help enable smartcities.

Demo model of the project where the nanogenerator installed on rooftop provides enough energy to power the entire smarthouse below it

Language-translating earbuds are here

If you are trying to order dinner or find a hotel abroad, do not let your poor grasp of foreign languages make you panic. An upcoming in-ear gadget can now translate speech like Babel Fish in *The Hitchhiker's Guide to the Galaxy*, or Universal Translator gadget in *Star Trek*.

The system, dubbed Pilot, comprises two earpieces to be worn by two people who do not speak the same language, and uses an app, so the duo can converse with ease. It is designed to work offline, so it would not incur data charges when used overseas. Both wireless earpieces can also be used by one person for listening to music.

While the first-generation device works only when speaking to someone wearing an earpiece, future generations could listen to everything happening nearby, so pairs of devices would not be needed.

The gadget was invented by Waverly Labs founder, Andrew Ochoa, who said he had the idea for it after meeting a French girl.



An accompanying app enables both earpieces to toggle between languages